



# Ransomware Forensic Simulation Dashboard

Educational DFIR Lab Analysis Report



## Attack Summary

Files Encrypted

13

Attack Duration

~1 minute

File Types

3 types (.txt, .docx, .jpg)

Ransom Note

YES ✓



## Encryption Timeline

|   | timestamp                  | filename                  | action                |
|---|----------------------------|---------------------------|-----------------------|
| 0 | 2026-07-13T01:29:13.832361 | README_RESTORE.txt        | create_encrypted_file |
| 1 | 2026-07-13T01:29:13.832361 | README_RESTORE.txt        | delete_original_file  |
| 2 | 2026-07-13T01:29:13.834224 | README_RESTORE.txt.locked | create_encrypted_file |
| 3 | 2026-07-13T01:29:13.834224 | README_RESTORE.txt.locked | delete_original_file  |
| 4 | 2026-07-13T01:29:13.835148 | annual_report.docx        | create_encrypted_file |
| 5 | 2026-07-13T01:29:13.835148 | annual_report.docx        | delete_original_file  |
| 6 | 2026-07-13T01:29:13.835678 | client_list.docx          | create_encrypted_file |
| 7 | 2026-07-13T01:29:13.835678 | client_list.docx          | delete_original_file  |
| 8 | 2026-07-13T01:29:13.835918 | company_logo.jpg          | create_encrypted_file |
| 9 | 2026-07-13T01:29:13.835918 | company_logo.jpg          | delete_original_file  |
|   |                            |                           |                       |



## Encrypted Files

|   | File Name                    | Original Name         | Size (bytes) | Status   |
|---|------------------------------|-----------------------|--------------|----------|
| 0 | annual_report.docx.locked    | annual_report.docx    | 120          | 🔒 Locked |
| 1 | client_list.docx.locked      | client_list.docx      | 120          | 🔒 Locked |
| 2 | company_logo.jpg.locked      | company_logo.jpg      | 100          | 🔒 Locked |
| 3 | family_photo.jpg.locked      | family_photo.jpg      | 100          | 🔒 Locked |
| 4 | journal.txt.locked           | journal.txt           | 120          | 🔒 Locked |
| 5 | passwords.txt.locked         | passwords.txt         | 120          | 🔒 Locked |
| 6 | quarterly_budget.docx.locked | quarterly_budget.docx | 100          | 🔒 Locked |
| 7 | safety_canary.txt.locked     | safety_canary.txt     | 100          | 🔒 Locked |
| 8 | screenshot.jpg.locked        | screenshot.jpg        | 100          | 🔒 Locked |
| 9 | server_ips.txt.locked        | server_ips.txt        | 100          | 🔒 Locked |
|   |                              |                       |              |          |

## Encryption Key

### Key Value

Pfb4A6IRxFbs2J7BEjh9FXAvtr2PsxHLxvQyjNsPLSQ=

### Key Info

Algorithm: AES-256 Key Length: 44 chars Format: Hexadecimal

## Detection Signatures

```
rule Ransomware_Sim_Detection {
  strings:
    $note = "README_RESTORE" ascii
    $ext = ".locked" ascii
  condition:
    any of them
}
```

## Key Findings

### Attack Characteristics

1. 22 files encrypted in ~1 minute
2. All files renamed with .locked extension
3. README\_RESTORE.txt ransom note created
4. Original files deleted after encryption

## Recommendations

- Implement File Integrity Monitoring (FIM)
- Deploy EDR solutions
- Maintain 3-2-1 backup strategy
- Network segmentation
- Monitor encryption patterns

 Educational DFIR lab in isolated VMware environment